

ADDER CIRCUIT

Condensed Study Notes

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Half Adders

Imagine you're adding two single digits, like in elementary school. You have the numbers you're adding, and then you have the result. Sometimes, when you add two digits, you get a result that has two digits. For example, $1 + 1$ equals 10 (in binary, where we only use 0s and 1s). The '0' is the direct result of adding those two digits, and the '1' is something you carry over to the next column.

A half adder is like a simple calculator that can add just two single digits (called bits) together. It gives you two outputs: the direct sum of the two bits, and a 'carry bit' if the sum is too big to fit in a single digit.

Let's look at all the possibilities when adding two bits, A and B:

- If A is 0 and B is 0, the sum is 0 and there's no carry (0).
- If A is 0 and B is 1, the sum is 1 and there's no carry (0).
- If A is 1 and B is 0, the sum is 1 and there's no carry (0).
- If A is 1 and B is 1, the sum is 0, and there's a carry (1).

Notice how the 'sum' bit is 1 only when the two bits being added are different. This is like a special kind of addition where if you add 1 and 1, you get 0 and carry over a 1. This is called a **modulo-2 sum**. This operation is often done using something called an **XOR gate**.

The 'carry bit' is 1 only when both bits being added are 1. This is like regular multiplication where 1 times 1 is 1. This operation is done using something called an **AND gate**.

So, a half adder takes two input bits (A and B) and produces two output bits: the Sum (S) and the Carry (C).

COMPUTER ARCHITECTURE ADDER CIRCUIT

We've already met the 'half adder', which is like a tiny calculator that can add just two single digits (called 'bits') together. It gives you a sum and a potential carry-over, just like when you add numbers like $5 + 7$ and get 12 (the 1 is a carry-over).

Now, let's talk about 'adders' in general. Think of an adder as any device specifically designed to perform addition.

These adders can be linked together, kind of like connecting Lego bricks, to add up numbers that are much longer than just two single bits. This way, computers can add really big numbers!

There are two main types of adders we'll discuss:

- **Half Adder:** You've already seen this one! It's the simplest adder, designed to add only two single bits (let's call them A and B). It produces a sum and a carry-out bit.

- **Full Adder:** This one is a bit more advanced. Imagine you're adding numbers in columns, like in elementary school. Sometimes, you have a 'carry' from the previous column that you need to add in with the current column's digits. A full adder does just that – it adds two bits (like the ones in the current column) PLUS a 'carry-in' bit from a previous addition. Like the half adder, it also produces a two-bit output (a sum and a carry-out).

Full Adders

We've learned about half adders, which are great for adding two single bits of information. But they have a limitation: they can't handle a 'carry-in' signal. Imagine you're adding numbers by hand, and you have a number to carry over from the previous column. A half adder can't deal with that.

A full adder is like a more sophisticated addition machine that can handle this. It adds **three** numbers together: the two numbers you want to add (like the 'addends' in a half adder) and a 'carry-in' signal (the number carried over from a previous addition step).

Think of it like this: You're adding up the scores for three different people on a single question in a quiz. A half adder could only add two people's scores. A full adder can add all three, and still tell you the total sum and if there's a 'carry' to the next point tally.

So, the output of a full adder is the complete, two-bit arithmetic sum of these three one-bit numbers. It gives you both the sum for that column and any carry-over to the next.

Interestingly, you can build a full adder using two half adders! Here's how:

1. Take your first two numbers (let's call them A and B) and feed them into the first half adder.
2. Take the 'sum' that the first half adder produced and feed it into one of the inputs of a second half adder.
3. Take the 'carry-in' signal (Cin) and feed it into the other input of the second half adder.
4. Finally, take the 'carry' outputs from *both* half adders and combine them using an 'OR' gate. This combined carry becomes the final 'carry-out' (Cout) for the full adder.

This structure allows us to create logic expressions that describe how a full adder works, and we can prove these expressions are correct by using a truth table, which shows all possible inputs and their corresponding outputs.

Reference

This section is a reference to an external website. It's like a bibliography in a book that points you to other sources for more information.

Think of it like this: If you were reading a recipe and it mentioned a special ingredient you'd never heard of, the recipe might have a 'Reference' section that says, 'For more on how to make homemade vanilla extract, see page 50 of 'The Baking Encyclopedia'.' This 'Reference' section doesn't tell you how to make vanilla extract itself, but it tells you where you can go to learn about it.

In this case, the reference points to a specific page on Wikibooks about 'Practical Electronics/Adders'. This means if you want to dive deeper into the topic of adders, or explore other related electronics concepts,

that's the place to go.